

A Guide to Computer Parts

I. INTRODUCTION

There are many reasons to build a computer: necessity, entertainment, curiosity, control, and even making connections with family. It's a fantastic activity and can be immensely satisfying to have hard work pay off in such a significant way. It also gives strict control over exactly what goes in the computer, so there is the utmost confidence that the computer is running with only the highest quality parts.

If only it were easy to find those quality parts. Manufacturers seem to make it their life's goal to make their products as confusing as possible to identify, information on parts can be surprisingly difficult to locate, and terminology can be excessively complicated. This all assumes that the manufacturer isn't being deceptive, which certain lower-end manufacturers have no qualms about.

This guide serves as an entry point: a tool to help navigate through the crazy world of computer parts. A tool to help instill confidence in buying decisions. A tool to encourage those who might be considering making a PC to call their own.

II. METHODOLOGIES WHEN PICKING PARTS

Before parts can be chosen, what is wanted must be determined. When picking parts, it is very important to consider why each part is being chosen and if it's appropriate for the user's use cases. There are four general categories to consider before picking any parts: Use case, budget, metrics, and special considerations.

I. Use Case

There are many different use cases for a personal computer, from playing video games to browsing the internet and everywhere in-between. The primary use case of the computer is what the computer is going to spend the most time doing. This should be the strongest factor when making any kind of part decision. Some examples of different use cases are listed below:

- A. Web browsing: ★
- B. Development: ★★
- C. Gaming: ★★ ~ ★★★★★
- D. Video editing: ★★★★★ ~ ★★★★★★
- E. Streaming: ★★★★★ ~ ★★★★★★
- F. Server hosting: ★★★★★
- G. 3D modeling: ★★★★★★

Other use cases do exist, and in most situations use cases are some combination of these. They can also be an end goal or a future consideration, like wanting to stream or learning how to edit videos. Parts often work for multiple use cases, so when building a PC for video editing it will also work well for gaming.

The stars represent the *relative* computing demands for each category. There is a large amount of variance in requirements, based on the use case. Make sure to be as granular as possible when defining use cases: for example, Cyberpunk 2077 is a much more demanding of a game than Minecraft, so playing Cyberpunk 2077 is going to take more powerful (and expensive) parts. This also means that any computer that can play Cyberpunk 2077 can also play Minecraft. For another example, editing higher quality videos is going to take more power, so part decisions must be made with that in mind.

II. Budget

Budget represents the amount of money that can be allocated to both the entire build and each part. Allocating more money will generally see a linear increase in overall performance for a computer, but there are diminishing returns around \$2,000 or more. This does not mean that getting a \$2,000 computer is a requirement; building a computer for web browsing does not require \$2,000. This is why use case is the first consideration, since the budget for parts should be dependent on what the computer is going to be used for.

When budgeting, try to find prices that are as close to the Manufacturer Suggested Retail Price (MSRP) as possible. Try to avoid budgeting around sales, as they can make purchasing things difficult based on wait times for a sale and stock considerations. If the plan is to wait until a large event, like Black Friday or Cyber Monday, it is best to stick with a generalized budget for the whole build and to pick parts based on the best available sales. It is also important to consider used sites, like eBay or Facebook Marketplace, that can sell used hardware at a much lower price. Used hardware is still good hardware and can be taken advantage of to save a significant amount of money.

As a note, this guide will not be covering pricing for any part. Pricing can fluctuate significantly and MSRP for products can increase if the demand for a specific part is high enough.

III. Metrics

While creating a budget for picking parts is important, the price of a part can often be deceptive when determining quality. As a result, this guide will be using *metrics* to define what is a quality part. These metrics are generally benchmarks done by independent reviewers, free from the influence of the manufacturers that make the products. Metrics also serve as the best way to determine how well a part performs, both relative to the competition and absolutely. The graphic below shows an example of a popular benchmarking software called Cinebench (Fig. 1):

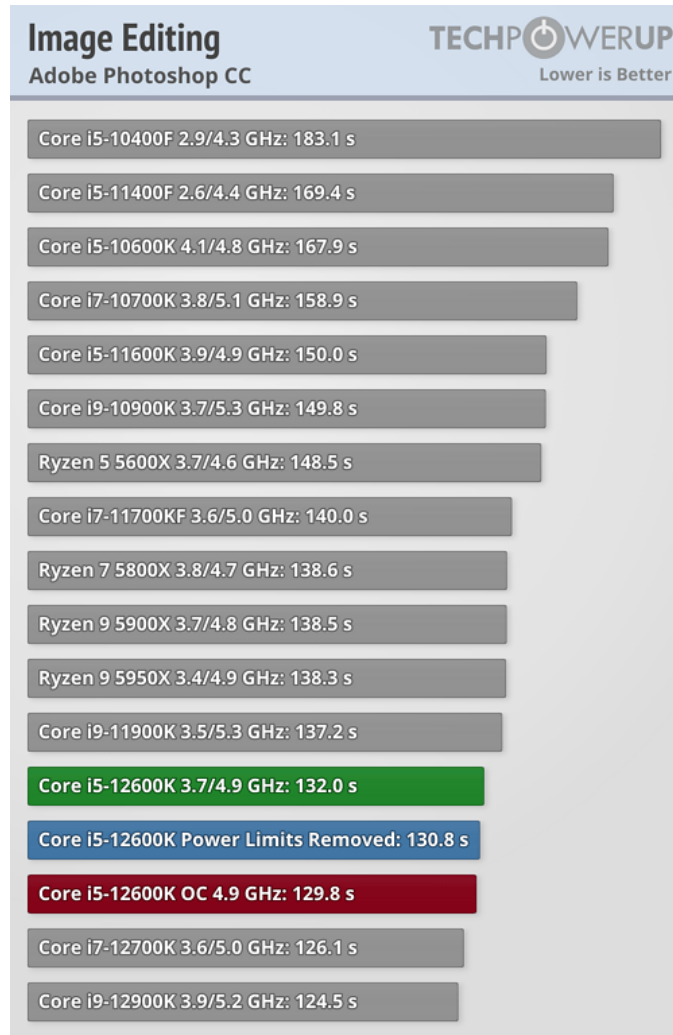


Fig. 1. Adobe Photoshop image editing times for Intel and AMD processors

Source: <https://www.techpowerup.com/review/intel-core-i5-12600k-alder-lake-12th-gen/11.html>

This graphic has a lot of information, but the important thing to note is the comparison between parts. In this example, the green bar represents the product being reviewed, and the other two bars represent the same product with certain modifications. Every other bar is a competing product, showing how good or bad that product is compared to the competition. These graphics will be placed when relevant to the discussion but will be used sparingly. When deciding what product to purchase, these reviews are going to be the best resource for determining what part to pick. The processor listed in green is a full \$300 cheaper than the processor at the very bottom (Intel) at MSRP! That is a very high price to pay for saving eight seconds when working in Adobe Photoshop.

There will be links in the resources section which contain reviews for products from trusted reviewers in the industry. Make sure to utilize them when shopping for parts.

IV. Special Considerations

In no particular order, here are some special considerations to make when picking hardware:

1. Operating systems are not included in this guide, primarily because they are software. The only real choices are the current version of Windows and a version of Linux like Ubuntu.
2. This guide focuses heavily on picking parts to build a custom computer, but the information contained within will be just as useful for somebody who wants to purchase a computer that has already been built for them. These are referred to as pre-built computers.
3. This guide is designed specifically to give the information needed to identify and make sense of the esoteric practices the computer hardware industry is known for. It is to help facilitate independent research, and no distinctions are made about the superiority of any given product. Parts and companies can change significantly over a relatively short period of time, so certain concessions must be made when trying to make a guide that can stand the test of time.
4. When making any kind of claims about time, the statements are based on the time of the guide's creation, December 2, 2021, and may not be correct in the future. Parts standards can change, as can prices, so it's important to keep this in mind.
5. This guide will not have any information on how to build a computer, just for finding parts for the computer. There will be some resources in the resources section of the document that contains tutorials on how to build a computer.

III. PROCESSOR (CPU)

In terms of the human body, the central processing unit (CPU) would be the brain. It does all the computational heavy lifting so that a computer can play a video on YouTube or send an email to grandma. A computer without a good CPU will perform slower, with applications taking more time to perform their necessary duties and some applications not being able to function at all. A faster CPU will see better performance throughout the system, from programming to playing video games.

When it comes to picking the CPU, there are two main choices: Intel and AMD. These two companies have fought each other for the title of best processor for over 30 years, and both companies have seen their fair share of wins and losses (Encyclopedia Britannica). The choice between the two companies is largely up to circumstance, depending on budget, use case, and metrics.

It is important to note that this decision will dictate what other parts you can purchase. Intel CPUs are only compatible with certain parts, and AMD CPUs are only compatible with certain parts. They are **not** cross-compatible.

I. Intel

Intel holds the noteworthy distinction of being the manufacturer of the world's first microprocessor, which set the stage for the more familiar CPUs we know today (Cass).

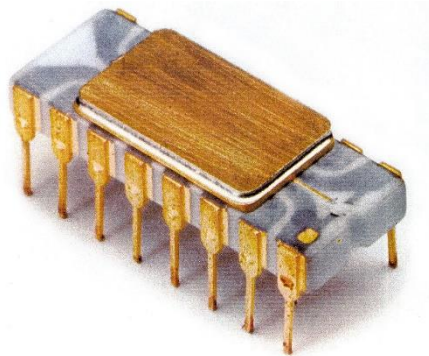


Fig. 2. Intel 4004 Microprocessor, the world's first microprocessor

Source: <https://spectrum.ieee.org/chip-hall-of-fame-intel-4004-microprocessor>

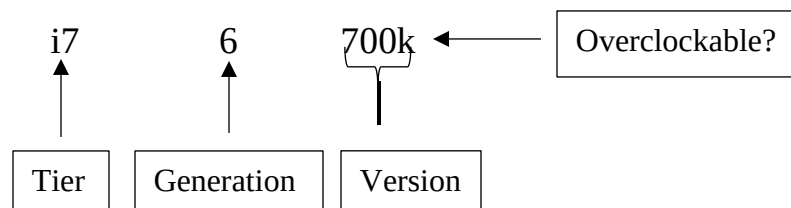
Nowadays, processors look very different. They also perform very differently, vastly outperforming the processors even five years prior. An example of a modern Intel processor is shown below.



Fig. 3 & 4. Front and back of an Intel i7-6700k Skylake processor

Source: [https://en.wikipedia.org/wiki/Skylake_\(microarchitecture\)](https://en.wikipedia.org/wiki/Skylake_(microarchitecture))

The front of the processor has a metal plate with some information on it that helps us identify what processor it is, and the back has many contacts that will connect to the motherboard. This information on the front can be a little confusing, but the most important piece of information on the picture is the model number (underlined in red). This can be broken down into four distinct parts:



The tier refers to how good Intel thinks the product is relative to the other groups of products within the same generation. The generation refers to the number of iterations that Intel has gone through when creating processors using this method. The version is a more granular metric than tier, indicating the level of features of the processor compared to the rest of the generation. The “k” simply represents if the processor can be overclocked, which is a nice feature but irrelevant for most people.

The generation, version, and the letter k (if one exists) are the most important values, as those will make it easier to search up the metrics for each part. All other Intel processors will follow a very similar naming format.

II. Advanced Micro Devices (AMD)

Starting in 1969, AMD and Intel have had a long and storied rivalry in the processor manufacturing space (Encyclopedia Britannica). AMD was the first company to pass the 1-GHz (gigahertz) threshold, and after a small period of underperformance in the market they have once again risen to prominence, overshadowing Intel until just this year.

Modern day AMD processors look like Intel processors and follow a lot of the same conventions that Intel does for naming.

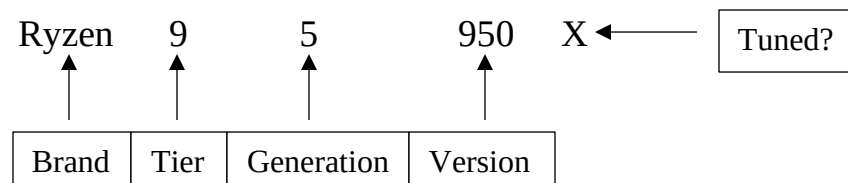


Fig. 5. Front and back of AMD Ryzen 9 5950X processor

Source: <https://www.forbes.com/sites/antonyleather/2021/07/29/amd-zen-4-cpu-and-rdna-3-gpu-2022-launches-confirmed-as-company-posts-impressive-revenue/>

The processor shown in the picture above shares a lot of similarities with the Intel processor, but there is one key difference: there are pins instead of contacts. This has a lot of important implications, but the two most important ones are that the bottom is more fragile, and the processor won't fit on parts that support Intel CPUs. This is the main reason for lack of compatibility between the two platforms.

The numbers underlined in red will once again help us identify what part this is:



The brand just represents the current product name since AMD has had multiple different branded products. The tier, generation, and version all represent the same thing as an Intel processor, but the numbers differ simply because of the number of products and number of iterations AMD has had compared to Intel. The “X” represents a processor that has been tuned slightly, offering slightly more performance than the non-x counterpart (if one exists).

The brand, tier, generation, version, and tuning are the most important values to keep track of, as they will make it easier to look up metrics for the product. All AMD products within the past four years use this naming scheme.

IV. MOTHERBOARD

If the CPU is the brain, then the motherboard is the nervous system. It is responsible for making sure all the different parts in a computer communicate with each other as efficiently as possible. Without the motherboard, the CPU would be unable to do much of anything, so some argue that it is the most important part in the computer.

The number of different motherboards is vast, and the options each board comes with can be overwhelming. Picking the CPU first makes this decision much easier, as it cuts the number of options available by about half. From there, there are three factors to consider when making a motherboard decision: socket, size, and ports.

I. Sockets and Chipsets

The socket refers to the space on the motherboard that is dedicated to the processor. Each motherboard has a specific socket, which will only support a specific range of processors. Functionally, this means that the motherboard chosen must be compatible with whatever processor was chosen, or vice versa. If the processor was picked first (which this guide is assuming), then the correct socket can be determined by looking at the specifications for the processor chosen. This is where the values discussed in the processor section of the guide become quite useful: the values can be used to locate the metrics related to the chosen processor:

Package Specifications	
Sockets Supported ?	FCLGA1151

Fig 6. Socket for Intel i7-6700k processor

Source: <https://ark.intel.com/content/www/us/en/ark/products/88195/intel-core-i76700k-processor-8m-cache-up-to-4-20-ghz.html>

CPU Socket:	AM4
--------------------	------------

Fig. 7. Socket for AMD Ryzen 9 5950X

Source: <https://www.amd.com/en/product/10456>

The two images above both show the sockets for each of the two processors talked about in the processors section of this guide. This means that any motherboard purchased *must* be the same socket as the one listed on the website for each company. Extra special care must be taken for AMD processors, as multiple generations of processors all use the same socket but not all motherboards support all the processors that use the same socket.

AMD Socket AM4 Motherboard Chipset Solutions							
	AMD Athlon™ Processors with Radeon Graphics	AMD Ryzen™ 1000 Series Processors	AMD Ryzen™ 2000 Series Processors with Radeon™ Graphics	AMD Ryzen™ 2000 Series Processors	AMD Ryzen™ 3000 Series Processors with Radeon™ Graphics	AMD Ryzen™ 3000 Series Processors	AMD Ryzen™ 5000 Series Processors
	Compatible with	Compatible with	Compatible with	Compatible with	Compatible with	Compatible with	Compatible with
X570	X	X	X	✓	✓	✓	✓
B550	X	X	X	X	X	✓	✓
A520	X	X	X	X	X	✓	✓
X470	✓	✓	✓	✓	✓	✓	Selective Beta BIOS update needed
B450	✓	✓	✓	✓	✓	✓	Selective Beta BIOS update needed
X370	X	✓	✓	✓	✓	Selective Beta BIOS update needed	X
B350	X	✓	✓	✓	✓	Selective Beta BIOS update needed	X
A320	✓	✓	✓	✓	✓	X	X

Fig. 8. Compatibility chart for AMD motherboards that support the AM4 Socket

Source: <https://www.amd.com/en/chipsets/b550>

Sockets are only part of the equation when ensuring compatibility with the processor, however. Multiple motherboards can have the same socket, but the chipset determines many of the features that a motherboard offers: number of available ports, compatibility with certain technologies, and even the number of allowed expansion cards (the graphics card is one example of an expansion card).

It can be very difficult to determine what chipset to choose, but the best way is to look at the chipset numbering; more recent chipsets will have a bigger first digit, and chipsets with more features will have larger numbers. For example, a B450 board is going to have less features than a B550 board because it is a generation older, and an X570 board is going to have more features than a B550 board. The same concept applies to Intel motherboards, but it is less complicated because each motherboard only supports one CPU generation instead of multiple like AMD does.

In all cases, it is best to pick the board which best matches the features you are looking for.

II. Sizes

Motherboards come in a variety of sizes, from very small to gigantic and everywhere in between. While this does have less significance on the raw features available for each motherboard, it does reduce the amount of expandability each motherboard has. Smaller motherboards will have less expansion slots and less ports to plug peripherals into, so it is important to pick a board size that can accommodate future expansion.

There are many different sizes, but the four most common ones are shown below:



Fig. 9. Four most common motherboard sizes

Source: <https://www.techosaurusrex.com/motherboard-size-comparison-chart/>

Micro-ATX and ATX are the most common sizes, with Mini-ITX being for extremely compact computers and EATX being for people who need a lot of expansion ports. The choice itself doesn't have too much influence on the performance of the computer, but it can drastically affect how many options for upgrades are available in the future. This also influences what case is chosen, as an ATX board won't fit in smaller cases.

III. Ports

Arguably the most pertinent for the user, the number and type of ports available on a motherboard can be a heavy influence on what motherboard is chosen. Each person has different needs, and it is important to pick a motherboard that accommodates those needs.



Fig. 10. Ports on ASUS Z690 PRIME Motherboard

Source:

https://www.newegg.com/p/N82E16813119507?Description=lga%201700&cm_re=lga_1700_-13-119-507_-Product&quicklink=true

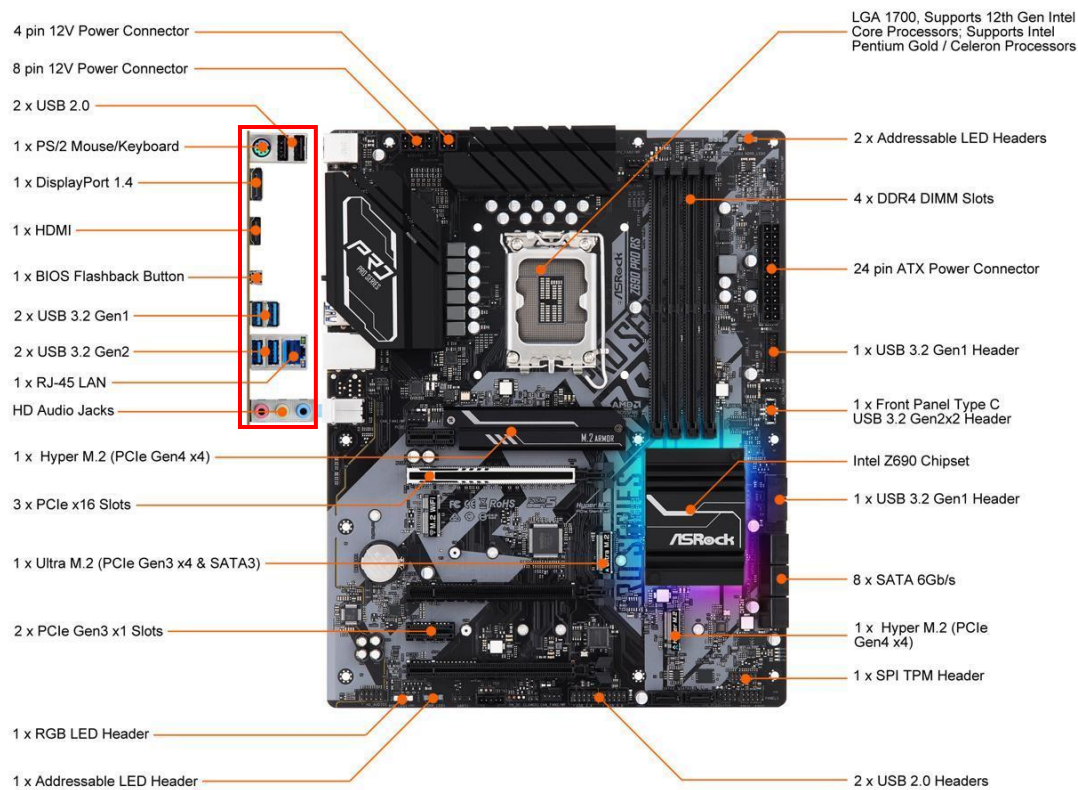


Fig. 11. Ports on ASRock Z690 Pro RS Motherboard

Source:

https://www.newegg.com/p/N82E16813162039?Description=lga%201700&cm_re=lga_1700_-13-162-039_-Product

There's a lot of information visually on these two screenshots, and it can be overwhelming, so we are going to focus on just the ports surrounded by the red boxes right now. They both look similar at first, but the top motherboard has a few distinct features the bottom motherboard doesn't have: it has faster networking with a 2.5GB ethernet port, two USB-C ports for high-speed connections, and more audio jacks for a surround sound setup. Not everybody needs those features, but it is important to some people so it might be a better option.

The motherboard shown on the top also has more slots available for a special kind of super-fast storage that the bottom board doesn't have. However, the motherboard shown on the bottom has more big expansion ports that support larger expansion cards, which can be useful for somebody who is intended to add more expansion cards in the future.

Ports may not seem important, but they can cause frustration over the long term if there isn't enough of them and having the right features can save on extra purchases in the future.

V. GRAPHICS CARD (GPU)

If the CPU is the brain of the computer, then the GPU, or graphics processing unit, is the eye. The graphics card deals almost entirely with displaying images to the user through the monitor, rendering the images and sending them to the monitor upwards of hundreds of times per second. The graphics card is actually very similar to a CPU but is designed from the ground-up to perform the task of displaying images as efficiently as possible.

There are two major companies which produce graphics cards: Nvidia and AMD. However, it is important to mention that most Intel processors come with their own graphics cards built into the processor. AMD processors do *not* generally have built-in graphics cards, otherwise known as integrated graphics. If the computer is being used for basic web browsing it may not be necessary to purchase a graphics card if an Intel processor was chosen.

When deciding which graphics card to go with, the best source of information is performance benchmarks. This is especially important for the people who wish to play video games, as a graphics card has a huge amount of influence on how well a video game performs.

I. Nvidia

Founded in 1993, Nvidia has spent the past 28 years creating and innovating in the graphics processing space (NVIDIA). They have long been a prominent player in the space, creating the first GPU and innovating on many technologies like real-time raytracing in video games (NVIDIA).

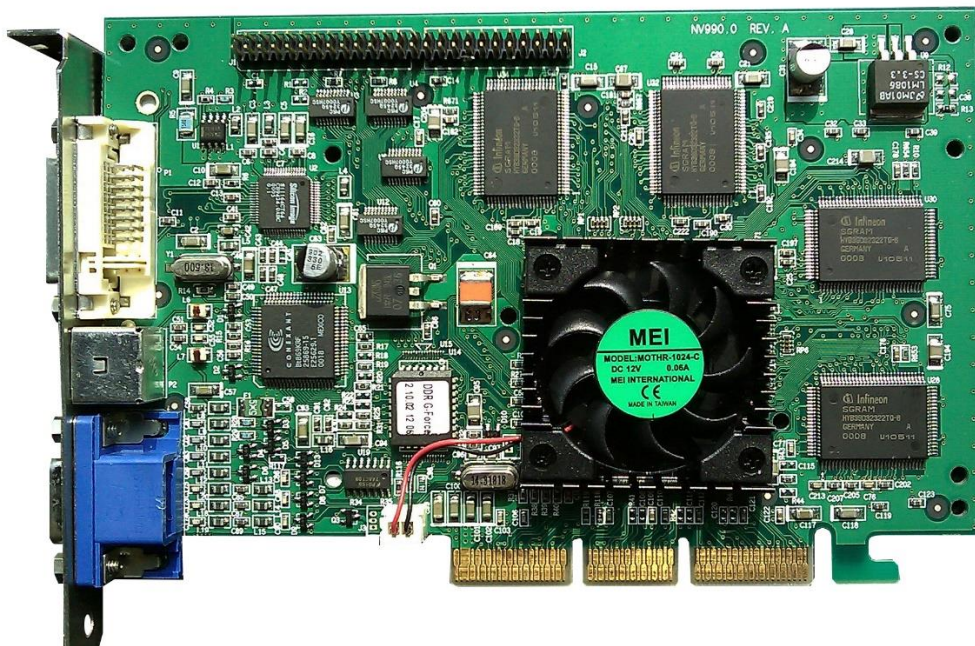


Fig. 12. Nvidia's first GPU

Source: https://en.wikipedia.org/wiki/GeForce_256

The image above shows an example of an old GPU, but modern graphics cards have changed a significant amount visually. They have increased in power and gotten larger as a result. They also usually come with hefty coolers attached to dissipate the heat that power generates. A modern Nvidia GPU is shown below:



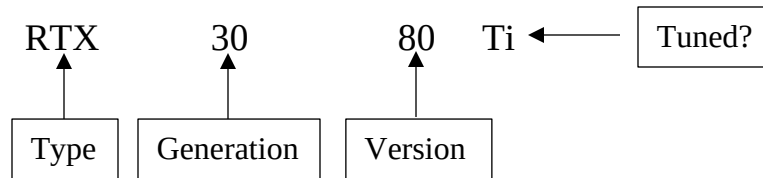
Fig. 12. EVGA RTX 3080 Ti

Source: https://www.newegg.com/evga-geforce-rtx-3080-ti-ftw3-ultra/p/1FT-0009-00712?Description=EVGA%20rtx%203080ti&cm_re=EVGA_rtx%203080ti--9SI AKRFGF35464--Product&quicklink=true

It has the same general structure, but is much larger in all significant aspects, being almost a foot long, five and a third inches tall, and 2.44 inches wide, this can be an important consideration when picking certain parts. For example, some cases don't have the clearance necessary to fit this card, and the width means that other expansion slots on the motherboard might be covered due to the card's size.

It is also noteworthy that this card doesn't have any significant branding by Nvidia but is instead branded with the EVGA logo. This is called an add-in board graphics card, or AIB, and it refers to any GPU that isn't made by Nvidia directly but contains Nvidia parts. Nvidia gives these companies the schematics and chips needed to build their own graphics cards, allowing them to tweak the performance and add better cooling. These cards are usually sold at a markup compared to what Nvidia sells them for as a tradeoff, but they are generally a quality product and perform indistinguishably (or better) compared to what Nvidia produces.

Nvidia GPUs follow a numbering format that is very similar to the Intel and AMD processor numbering scheme:



The most important values are the generation, version, and whether it's "tuned" or not. The type doesn't mean much of anything and is there specifically for marketing. The generation represents the current iteration of graphics cards, but Nvidia abandoned that once it hit the number 10 and has been going up by 10s ever since, so this is technically their 13th generation of cards, not the 30th. The "Ti" stands for titanium and is a fancy marketing way of saying that the card performs better than its non-Ti counterpart.

II. Advanced Micro Devices (AMD)

AMD doesn't just produce processors; they are also the chief competitor to Nvidia in GPUs. AMD made their entrance into the GPU space in 2006 with the purchase of technology company ATI, releasing their first series of GPUs in 2007 (AMD, "History of AMD").



Fig. 13. ATI Radeon HD 2400 XT

Source: https://en.wikipedia.org/wiki/Radeon_HD_2000_series

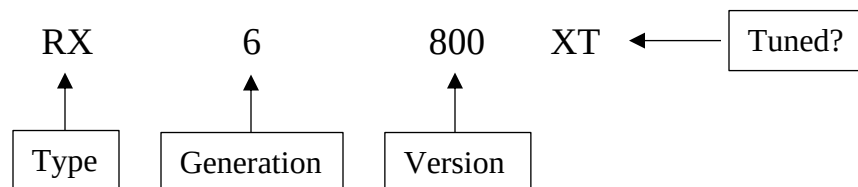
Just like Nvidia, much has changed from their older GPUs, like the one shown in the image above. They are much bigger, have much more cooling, and are much more power hungry.



Fig. 14. AMD Radeon RX 6800 XT

Source: <https://www.pcgamer.com/the-best-graphics-cards/>

All the same things that were mentioned in the Nvidia section above about modern day graphics cards apply to AMD parts: they have the same add-in board options from third-party companies, they can be up to a foot long in length, and they carry the same compatibility concerns with both cases and motherboards. They even follow the same naming scheme:



Just like the Nvidia cards, the type is primarily for marketing. The generation, version, and “tuned” label are the important values to take note of, and they all mean the exact same thing as the Nvidia graphics cards. Some caution must be taken with the generation, however, as AMD has not kept their naming schemes consistent for as long as Nvidia has. There are cards with larger numbers than this card but are much older, so when

searching around make sure to include the type in the search to ensure the right card is found.

VI. MEMORY (RAM)

The memory, or RAM, serves as the short-term memory of a computer. A processor (just like a brain) can't really do much calculating if it doesn't remember what to calculate. The RAM, just like short-term memory, holds information about current operations going on throughout a computer. Computers without enough RAM will often see applications start to perform very slowly as the computer has to decide what operations to stop to free up more space (much like trying to have multiple important conversations at once).

Just like motherboards, there are a significant number of options available for memory. However, since the motherboard and processor both dictate what is compatible, the actual options available are much smaller. There are two factors to consider when looking at memory: type and speed.

I. Type

While historically there have been many different types of RAM, the current market is dominated by one singular type: DDR, or Double Data Rate RAM. When purchasing memory, all current motherboards and processors will support some variations of the DDR standard. The processor's specification sheet contains this information to help determine what type of DDR memory is compatible with the processor:

System Memory Type:	DDR4
---------------------	------

Fig. 15. Supported system memory type for AMD Ryzen 9 5950X processor

Source: <https://www.amd.com/en/product/10456>

Memory Types ?	DDR4-1866/2133, DDR3L-1333/1600 @ 1.35V
----------------	---

Fig. 16. Supported system memory types for Intel i7-6700K processor

Source: <https://ark.intel.com/content/www/us/en/ark/products/88195/intel-core-i76700k-processor-8m-cache-up-to-4-20-ghz.html>

For the AMD processor, only one type of system memory is supported, which is DDR4. The Intel processor supports two types: DDR4 and DDR3L. The 4 represents the generation, with bigger numbers being more recent generations. The L refers to RAM modules for laptops, so any RAM with the L in the name means it will only work with laptop systems. Motherboards that are compatible with the chosen processor will also be compatible with the same RAM that is supported by the processor. This also means that previous generations of DDR RAM are **not** compatible, and vice-versa.

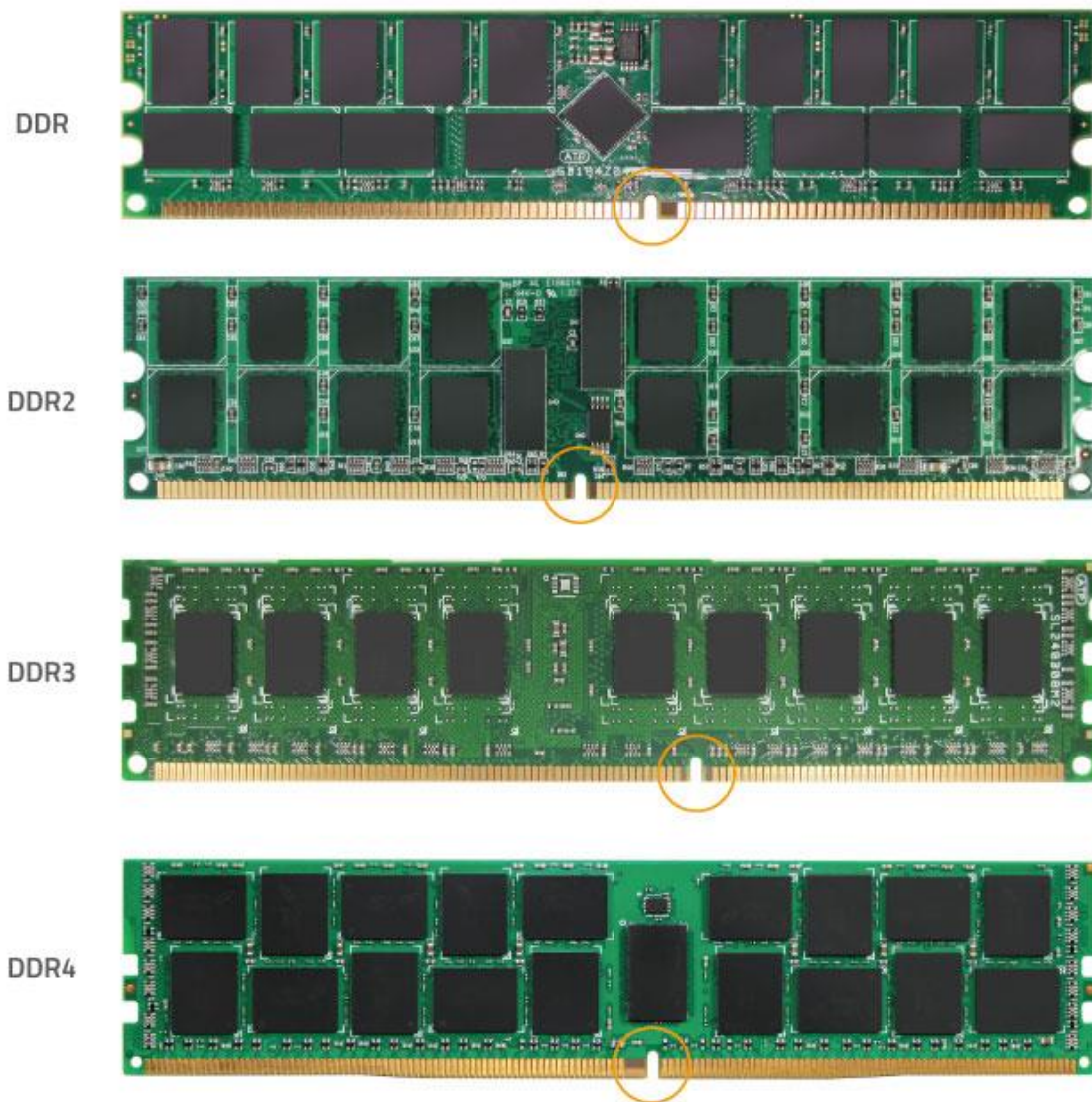


Fig. 17. Notch and pin differences in different standards of DDR memory
Source: <https://www.atpinc.com/de/blog/ddr4-vs-ddr3-differences-advantages>

As shown in the image above, the different generations of DDR all have notches in different locations to prevent accidental insertion of the wrong type of memory into a motherboard. Make sure to consult the processor's specifications page to ensure the right type of memory is selected.

II. Size and Speed

Once the type of RAM needed is determined, the size and speed can be chosen to best fit the use case of the computer. The size of each RAM module will determine how much memory a computer must work with overall, while the speed determines how fast the processor can retrieve and store information in the RAM. The number of modules any computer can have are based on both the number of slots available on the motherboard and the maximum amount of memory the processor can support. Most ATX motherboards have four slots for memory and most Mini-ITX boards only have two slots, for example. Size is additive, but speed is not. For example, if a computer has two 8GB (gigabyte) modules of memory, then a computer has a total of 16GB of memory to work with. If a computer has two modules of memory, each with a speed of 3200MHz (megahertz, or millions of cycles per second), then the maximum communication speed is still only 3200MHz.

Sizes and speeds are going to differ depending on the DDR generation that is compatible with the processor and motherboard. Conveniently, the processor's specifications sheet also lists the maximum supported speed:

System Memory Specification:	Up to 3200MHz
------------------------------	---------------

Fig. 18. Supported maximum memory speed for AMD Ryzen 9 5950X processor

Source: <https://www.amd.com/en/product/10456>

Memory Types ?	DDR4-1866/2133, DDR3L-1333/1600 @ 1.35V
----------------	---

Fig. 19. Supported maximum memory speeds for Intel i7-6700k processor

Source: <https://ark.intel.com/content/www/us/en/ark/products/88195/intel-core-i76700k-processor-8m-cache-up-to-4-20-ghz.html>

For the AMD processor, the maximum supported speed is 3200MHz. The Intel processor is a bit trickier, listing two different speeds: the first speed is if there are two or more RAM modules installed, and the second speed is if there is only one module installed. When purchasing RAM, it's generally advised to get the fastest speed supported by the processor, as this will increase performance.

Processors also have a maximum amount of memory that they can interact with. For the Intel i7-6700K, it is 64 GB (Intel) and for the AMD Ryzen 9 5950X it is 128 GB (AMD, "Processor Specifications"). This means that adding memory that exceeds the total capacity supported by the processor will be ignored, so it is important to keep in mind when deciding how much RAM is needed.

VII. POWER SUPPLY (PSU)

The power supply, or PSU, is arguably the most important part in the computer. It is the heart of a computer, supplying the necessary energy for all the other components to perform their duties without difficulty. A properly functioning power supply will have no visible impact on performance, but a poorly functioning power supply will wreak havoc, causing the computer to shut down randomly, suddenly crash, or even cause permanent damage. It is no exaggeration to say that the power supply is the lifeblood of any computer.

Unfortunately, power supplies are often some of the most confusing components to shop for. There is no surefire way to determine the quality of a power supply at a glance, so extra research must be done to verify that any given PSU is of sufficient quality. There will be some resources in the Resources section of this guide that can provide recommendations for quality power supplies that have been sufficiently tested. Instead, this section of the guide will focus on the information needed to choose an appropriate power supply.

As a note for those who are trying to stay within a specific budget: do **not** cut costs on a power supply to save a few dollars. It is never worth the risk of losing components, and it is never worth the heartache of not knowing why the computer keeps randomly turning off.

I. Wattage Requirements

All the components we have looked at so far all have one thing in common: they all need power. This power comes from the humble power supply, but not all power supplies are created equal. Every power supply is going to be rated for a certain wattage, listed somewhere on the product page and in the model number of the power supply.



Fig. 20. Seasonic TX-1000 1000W Power Supply

Source: <https://seasonic.com/prime-tx#gallery>

The power supply shown in the image above, the Seasonic TX-1000, has a maximum sustained output of 1000 watts (hence the 1000 in the model name). That means that this PSU can supply up to one kilowatt of power consistently for the lifetime of the power supply. Any power demands that exceed that total output can cause the computer to shut down as the computer is no longer getting enough power to continue functioning. This applies to all parts in the computer combined, so the processor, motherboard, graphics card, and memory all consume a portion of the total output.

This is only half of the equation, however; how is the total amount of power consumed by a computer determined? While certain parts list how much power they consume, and graphics card product pages often list a recommended minimum wattage, the best way to calculate the number of watts any system needs is a calculator:

CHOOSE YOUR COMPONENTS

The image shows a web-based wattage calculator interface. On the left, under the heading "CHOOSE YOUR COMPONENTS", there are four sections for selecting hardware:

- CENTRAL PROCESSING UNIT (CPU):** AMD (dropdown) and Ryzen 9 5950X (dropdown).
- MOTHERBOARD:** ATX (dropdown).
- GRAPHICS PROCESSING UNIT (GPU):** NVIDIA (dropdown), GeForce RTX 3080 Ti (dropdown), and a quantity of 1 (dropdown).
- RANDOM ACCESS MEMORY (RAM):** 16GB DDR4 (dropdown) and a quantity of 1 (dropdown).

On the right, a callout box displays the result: "Your recommended PSU wattage is: **581 Watts**". Below this are two buttons: "RESET" and "SHOP PSU". A small note at the bottom right states: "NOTE: The PSU wattage we recommend only gives you a general idea of what to consider when selecting a power supply. PCI cards, external devices, USB and FireWire devices, cooling fans, and other components may need more power."

Fig. 21. Wattage calculator by Newegg

Source: <https://www.newegg.com/tools/power-supply-calculator/>

These wattage calculators will supply a recommended wattage for a power supply based on the components chosen from the left side of the menu. These calculators aren't foolproof, but can serve as a great starting point to ensure that the power supply chosen will be capable of supplying the necessary power for components. When using calculators like this, it's best to add a couple hundred watts as a buffer to ensure consistent, smooth operation of the computer. For example, the image above recommends 581 watts, so looking for an 850-watt power supply would be a good choice. If an 850-watt power supply can't be found, moving up to the next common wattage is better than moving down.

II. 80Plus Ratings

The astute reader may have noticed the silver logo on the left side of the box on the power supply image above. This is called an 80Plus rating, and it serves to determine the efficiency of a power supply at a glance. Since computer parts use direct current (DC) and the power that comes out of the wall outlets is alternating current (AC), the power supply must convert from the AC coming out of the wall into the DC that the computer needs to function. This process takes power to do, so some of the power is lost when converting from AC to DC. The amount of power lost is measured as efficiency, with higher percentages meaning that less power was lost during the conversion. Higher efficiency power supplies are better both for saving costs on electric bills and providing better power for components.

The 80Plus badge means that the power supply always has an efficiency rating of at least 80%. The rating system for 80Plus is shown below:



	Loading	80 Plus	Bronze	Silver	Gold	Platinum	Titanium
Efficiency	20%	80%	82%	85%	87%	90%	90%
	50%	80%	85%	88%	90%	92%	92%
	100%	80%	82%	85%	87%	89%	94%

Fig. 22. Needed efficiency levels for each 80Plus certification

Source: <https://www.velocitymicro.com/blog/what-is-psu-efficiency-and-why-is-it-important/>

The lowest rating is just 80Plus, and the highest rating is 80Plus Titanium. The “Loading” column refers to how much of the power supply’s maximum power output is being used, with each tier being the minimum efficiency rating needed to meet that tier. For all ratings except the Titanium and the plain 80Plus, the highest efficiency rating is achieved when the load is at 50%, or around half of the maximum output of a power supply. This is the other reason why it is recommended to add extra wattage to the calculator when searching for a power supply: it increases the amount of time that the power supply in a computer is running at peak efficiency.

A word of caution for buyers: while these labels do guarantee a certain level of efficiency, they do not guarantee a certain level of quality. A power supply with an 80Plus rating is not guaranteed to be good, even if it is efficient. Make sure to do extra research on any power supply before purchasing one.

VIII. CPU Cooler

The processor can often run very, very hot. It is not uncommon for processors to hit temperatures more than 176 degrees Fahrenheit! If left unchecked, processors eventually get so hot that they damage themselves and can no longer function. This is where CPU coolers come in, doing exactly what the name describes. They help draw heat away from the processor and out into the world, keeping the CPU running in tip-top shape.

Unlike most other computer parts, CPU coolers are broadly compatible, working with most processors and motherboards across AMD and Intel without much effort. Of course, a full list of supported sockets is often listed on the product page, so that can be used to confirm if any CPU cooler is compatible.

There are two major types of CPU coolers: air coolers and liquid coolers. Both serve the same essential purpose, but they differ in implementation so certain considerations need to be made. They are usually about equivalent in terms of effectiveness, so picking between air cooling and liquid cooling is often a matter of preference.

I. Air Cooling

Air cooling, as the name implies, cools the processor using air. It is a bit more complicated than that, but the important thing to note is that air cooling is dependent on airflow to maintain peak cooling performance. While this method of cooling works well for most environments, there are certain considerations that must be made when picking an air cooler. Dusty environments can cause performance degradation over time, even damaging parts if left unchecked for too long, so regular maintenance will be required to maintain peak efficiency.

Many air coolers are quite large, taking advantage of surface area to efficiently dissipate heat away from the processor. This is another consideration that must be made when picking an air cooler or a case, since picking one first will require accommodation of the other.



Fig. 23. Cooler Master Hyper 212 Evo CPU Cooler

Source: <https://www.coolermaster.com/catalog/coolers/cpu-air-coolers/hyper-212-evo/>

The air cooler shown in the image above is a very popular model of air cooler, and it towers over the other components at 6.3 inches vertically. This means that some cases may not actually close properly with this cooler, which can look aesthetically displeasing and can be a safety hazard. It is also a very powerful cooler that can keep most processors cool with no issues.

Some processors come with an air cooler in the box, which is a great option for those looking to save some money on a build. It won't cool nearly as well, and isn't recommended for higher end processors that demand more power, but if the air cooler comes in the box it is sufficient for keeping the processor it came with cool. An Intel stock cooler is shown below:



Fig. 24. Stock cooler for Intel processors

Source: <https://www.amazon.com/Intel-Cooler-LGA1150-1155-E97378-001/dp/B00BQ1C4SS>

II. Liquid Cooling

Liquid cooling, as the name also implies, uses liquid (usually water) to transfer heat from the processor to a radiator, which dissipates the heat through fans. This means that liquid coolers are usually quieter than their air-cooling counterparts and can be less susceptible to dust issues. Some water coolers also look really cool and can add a lot of visual flair to any computer build.



Fig. 25. Corsair liquid cooled gaming computer

Source: <https://www.forbes.com/sites/antonyleather/2016/10/31/how-to-build-a-liquid-cooled-gaming-pc/?sh=5e2c7add5b46>

The computer shown above looks very impressive, with a red and white theme, red dyed liquid, and transparent tubing to show that liquid in motion. This build is also very expensive and uses very high-end parts that aren't practical for most users. This isn't intended to discourage doing so, just that it often takes a lot of work and can be prohibitively expensive.

The liquid coolers that most people will end up using is called an all-in-one liquid cooler, or an AIO. It looks like this:



Fig. 26. NZXT 120mm All-in-One liquid cooler

Source: <https://www.tomshardware.com/reviews/best-cpu-coolers,4181-2.html>

This is a more typical liquid cooler. It has all the basics of a liquid cooler: a pump, tubing, and a radiator. The pump, which is inside the cylinder with the purple NZXT logo on it, goes on top of the CPU. Warm liquid is moved from the inside of the pump to the radiator through the tubing, where the radiator helps dissipate the heat from the liquid using the attached fan. It is then pulled back into the pump, where the process repeats, keeping the processor cool. The 120mm measurement is the size of the radiator shown in the picture above, but radiator sizes can go up to 360mm. Keep in mind that the radiator must be attached to the case, along with fans, so making sure that the case supports the correct radiator size is important to ensure compatibility.

There is one special consideration to be made when picking a liquid cooler: liquid. The most common liquid contained within these coolers is water, and water doesn't play very nicely with electronics. The chances of failure are exceedingly low, but leaks do occur and can permanently damage parts.

IX. STORAGE

After a long day of word processing, gaming, and video editing, it's nice to be able to save that progress and resume the next day. This is what storage is for; it saves the work and holds applications so they aren't lost when turning off the computer. Storage serves as the long-term memory of the computer, preserving the state of the computer so it can be referenced when necessary. Photos, videos, word documents, spreadsheets, applications like a web browser, and many more things all reside in storage.

Storage comes in two distinct forms: solid state storage and mechanical storage. These are generally referred to as SSDs (solid state drive) and HDDs (hard disk drive) respectively. HDDs are larger in storage size (meaning they can hold more data) but operate slowly. The inverse is true of SSDs, where they are very fast but have comparatively less storage. Many people will have both HDDs and SSDs in their system to take advantage of the benefits of both.

I. Mechanical Storage (HDDs)

Mechanical storage has existed for a very long time, with the first commercial hard disk drive being shipped in 1956 (Computer History Museum):

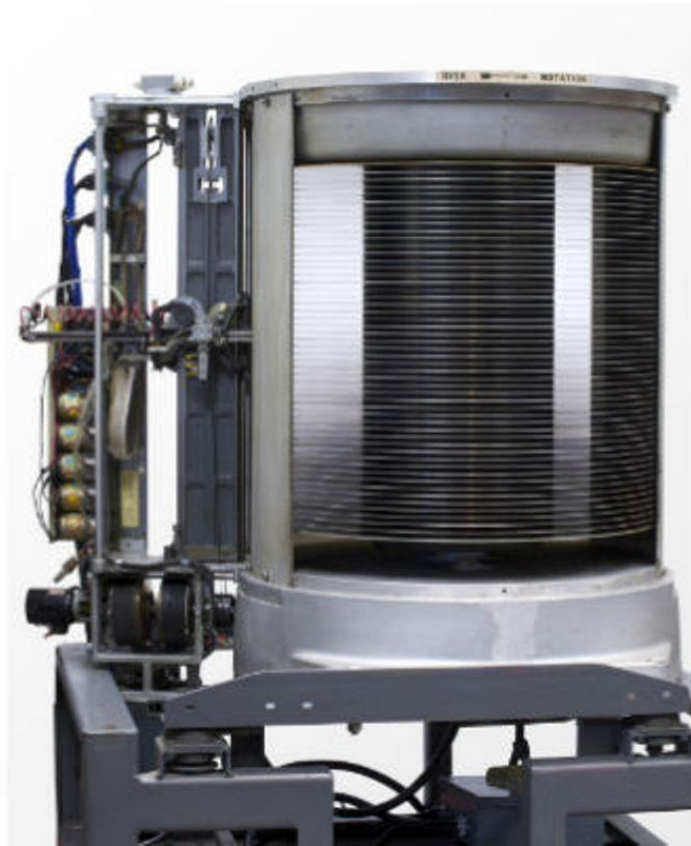


Fig. 27. First commercial hard disk drive ever shipped

Source: <https://www.computerhistory.org/storageengine/first-commercial-hard-disk-drive-shipped/>

Modern hard disk drives have gotten much smaller over the years, but they work in much the same way as the behemoth shown in the above image. A bunch of mechanical disks spin very quickly inside a metal chassis, with an articulating read-write head that goes over the disks as they spin and read/write data to the disk. These disks spin incredibly quickly, sometimes up to 15000 rpm. They also hold a large amount of data, so they are very useful for people who like to take lots of pictures and videos. A picture of a modern hard drive is shown below:



Fig. 28. Opened hard disk drive

Source: <https://acsdata.com/how-hard-drives-work/>

These drives are widely compatible with all the parts listed before this section. The only consideration that needs to be made is the case, since many modern cases are reducing the number of slots available for these bigger drives in favor of the more compact solid-state drives. If a large amount of storage is needed, make sure to pick a case that supports more hard disk drives.

II. Solid State Storage (SSDs)

Solid state storage is a more modern type of storage that doesn't use any mechanical parts. It relies on specialized chips that are designed to hold data even when the computer is turned off, meaning that these drives are not held back by the speed limitations of mechanical parts (like the disk rotating or the actuation head moving). This means that solid state storage operates much quicker, with less heat, and in a much more compact format. However, these do have a cost difference, and generally aren't available in the capacities that hard drives offer.



Fig. 29. Opened solid state drive

Source: <http://blog.superuser.com/2011/02/10/the-modern-marvel-of-the-ssd/>

A drive like the one shown above has the same connectors that modern hard disk drives do, so any part that supports a hard disk drive (just about all of them) also supports a solid-state drive. Most modern cases also have dedicated slots for an SSD, but this can vary so make sure to verify on the product listing.

There is another, cutting edge drive that just recently hit the market called an M.2 drive. These drives boast even faster speeds compared to even SSDs and come in an even smaller package.



Fig. 30. Intel 760p series M.2 solid state drive

Source: <https://en.wikipedia.org/wiki/M.2>

As the above picture shows, the drive itself is very small, only 3.14 inches long. It also has a special connector that is unique to it, which allows it to achieve those faster speeds compared to the other two types of drives listed. Since this drive model is so new, not all motherboards support it, so make sure to check the motherboard specification sheet to verify that M.2 drives are supported.

X. COMPUTER CASE

Fittingly, the last section of this guide covers computer cases, which wrap up all the previously discussed parts into one big box. “Box” doesn’t really do it justice, however; there are many different computer cases, from understated to extravagant:



Fig. 31 & 32. Examples of an extravagant case and an understated case

Sources: <https://www.corsair.com/us/en/Categories/Products/Cases/Mid-Tower-ATX-Cases/4000D-Tempered-Glass-Mid-Tower-ATX-Case/p/CC-9011198-WW>
& <https://www.newegg.com/p/N82E16811112533>

This is the best opportunity to really pick for style and show off those parts that took time and research to find.

The different sections of the guide all mention various considerations that must be made when picking a case. They are all explained in further detail in each section, but for convenience will be listed here:

1. Motherboard: Case must support the size of any chosen motherboard
2. Graphics Card: Case must support horizontal length of the graphics card
3. CPU Cooler:
 - a. Case must have a vertical clearance higher than the height of the air cooler
 - b. Case must support the radiator size chosen
4. Storage
 - a. Case must have enough hard disk drive bays to support however many hard disk drives purchased
 - b. Case must have enough solid-state drive bays to support however many solid-state drives purchased

Not all cases are created equally. Just like with all other parts, some cases are better than others, so it is a good idea to look at case benchmarks to determine how well a case performs. Performance in this case refers to airflow, with higher performing cases having better airflow. A case with poor airflow makes it difficult for the parts inside to get fresh air, causing the air inside the case to continually heat up and cause the temperature of the entire system to rise. The best CPU cooler is only as good as the case it is in. An example of some case benchmarks are below:

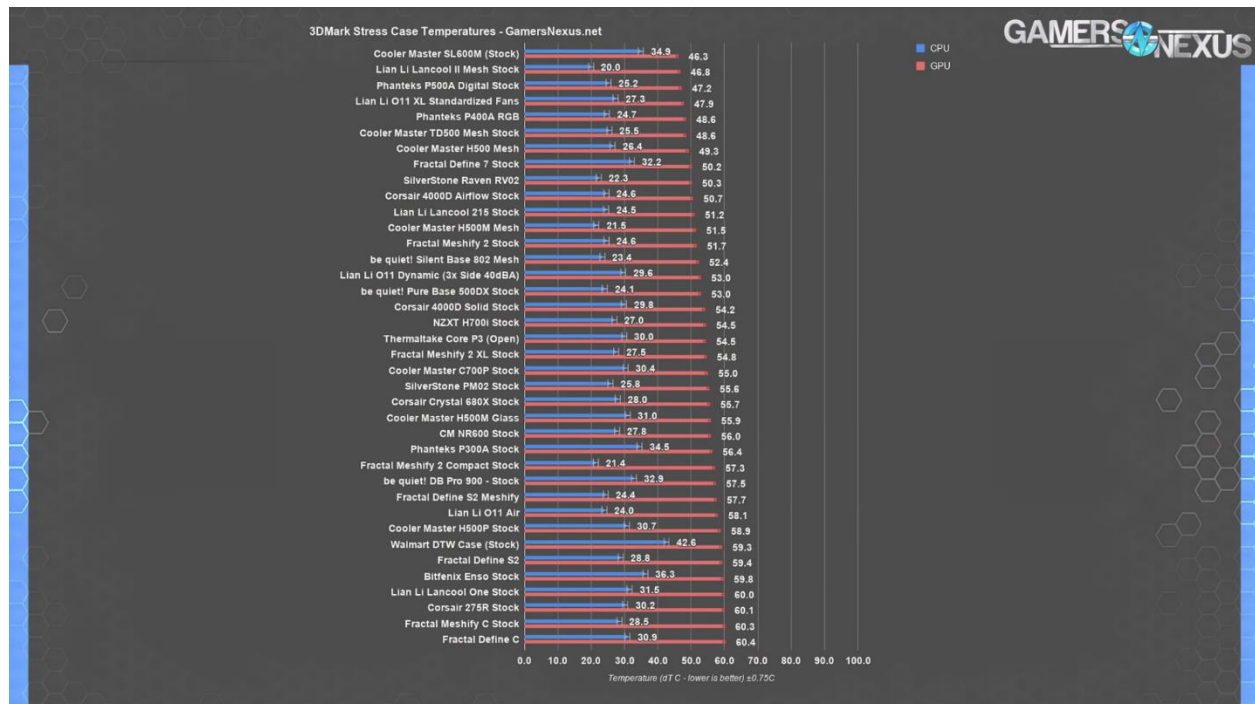


Fig. 33. Thermal ratings for large selection of cases when system is running a medium workload

Source: <https://youtu.be/ffuAnwGivO8?t=772>

The CPU and GPU temperatures for the cases can vary up to 14 degrees Celsius for the graphics card and a staggering 22 degrees Celsius for the CPU (that's 71 degrees Fahrenheit). These tests were taken in the same room, with the same room temperature, and all the same parts to show how much a case can have an impact on the temperature of the components inside.

XI. RESOURCES

This guide attempts to address the confusing nature of how parts are named and what parts are needed for a computer, but it doesn't cover many of the other aspects of computer building. Building the computer, finding quality reviews and benchmarks, and other issues that appear when trying to build a computer from scratch can be just as important. This section exists to provide links to resources that can help address some of these other aspects of this process.

The resources will be divided up by parts, just like the guide itself. There will also be a general section for non-specific resources.

I. General

LinusTechTips - [First Person View PC Build Guide](#)

PCPartPicker – [Website to assist in picking, pricing, and checking compatibility of parts](#)

LinusTechTips – [Trusted Independent Components Reviewer](#)

GamersNexus – [Trusted Independent Components Reviewer](#)

TechPowerUp – [Trusted Independent Components Reviewer](#)

II. Processor (CPU)

GamersNexus – [Best CPUs of 2021](#)

III. Motherboard

IV. Graphics Card (GPU)

V. Memory (RAM)

VI. Power Supply (PSU)

Newegg – [Power Supply Wattage Calculator](#)

LinusTechTips Forum – [Power Supply Tier List](#)

VII. CPU Cooler

GamersNexus – [Best CPU Coolers of 2021](#)

VIII. Storage

IX. Cases

GamersNexus – [Best Cases of 2021](#)

XII. CONCLUSION

Building a computer is an immense amount of fun, and it brings a lot of satisfaction in knowing that the work put in can bear such sweet fruit. This guide serves to lighten the load, giving readers the information they need to make more informed decisions about the parts their computer will be made of.

Works Cited

- AMD. "History of AMD." *AMD*, 3 Sept. 2007,
web.archive.org/web/20071012174217/http://www.amd.com/us-en/Weblets/0%2C%2C7832_10554%2C00.html?redir=AAMD04.
- . "Processor Specifications." *AMD*, 2021,
www.amd.com/en/products/specifications/processors.
- Cass, Stephen. "Chip Hall of Fame: Intel 4004 Microprocessor." *IEEE Spectrum*, 22 Oct. 2021,
spectrum.ieee.org/chip-hall-of-fame-intel-4004-microprocessor.
- Computer History Museum. "1956: First Commercial Hard Disk Drive Shipped | The Storage Engine | Computer History Museum." *Computer History Museum*,
www.computerhistory.org/storageengine/first-commercial-hard-disk-drive-shipped.
Accessed 5 Dec. 2021.
- Encyclopedia Britannica. "Advanced Micro Devices (AMD) | History & Facts." *Encyclopedia Britannica*, www.britannica.com/topic/Advanced-Micro-Devices-Inc. Accessed 5 Dec. 2021.
- Intel. "Intel Product Specifications." *Intel*, 2021, ark.intel.com/content/www/us/en/ark.html.
- NVIDIA. "NVIDIA Company History: Innovations Over the Years." *NVIDIA*,
www.nvidia.com/en-us/about-nvidia/corporate-timeline. Accessed 5 Dec. 2021.